

monitoring

* A stimulant such as d-amphetamine or methylphenidate (with caution; may activate bipolar disorder and suicidal ideation; may elevate blood pressure)

- Lithium
- Mood-stabilizing anticonvulsants
- Atypical antipsychotics (with special caution for those agents with monoamine reuptake blocking properties, such as lurasidone, ziprasidone, and zotepine)

Tests

- Patients should be monitored for changes in blood pressure; check before and 45–60 minutes after dosing until stable
- For patients receiving high doses or long-term treatment, consider periodically evaluating hepatic function

Side effects

How Drug Causes Side Effects

- Theoretically due to increases in monoamines in parts of the brain and body and at receptors other than those that cause therapeutic actions (e.g., unwanted actions of serotonin in sleep centers causing insomnia, unwanted actions of norepinephrine on vascular smooth muscle causing hypotension, etc.)

- Side effects are generally immediate, but immediate side effects often disappear in time

Notable Side Effects

- Agitation, anxiety, insomnia, weakness, sedation, dizziness
- Change in appetite, weight gain
- Sexual dysfunction (less than with other MAOIs such as phenelzine)
- Orthostatic hypotension (dose-related); syncope may develop at high doses



Life-Threatening or Dangerous Side Effects

- Hypertensive crisis (when MAOIs are used with certain tyramine-containing foods or prohibited drugs)
- Induction of mania in patients with bipolar disorder
- Rare seizures
- Rare hepatotoxicity

Weight Gain



- Occurs in significant minority

Sedation



- Many experience and/or can be significant in amount, especially daytime somnolence from nighttime insomnia
- Can also cause activation

What to Do About Side Effects

- Wait
- Wait
- Wait
- Lower the dose
- Take at night if daytime sedation; take in daytime if overstimulated at night
- Switch after appropriate washout to an SSRI or newer antidepressant

Best Augmenting Agents for Side Effects

- Trazodone (with caution) for insomnia
- Benzodiazepines for insomnia

Dosing and use

Usual Dosage Range

- 30–50 mg/day in divided doses

Dosage Forms

- Tablet 10 mg

How to Dose

- Initial 30 mg/day in divided doses; after 2 weeks increase by 10 mg/day each 1–3 weeks; maximum 60 mg/day, rarely up to 120 mg/day



Dosing Tips

- Orthostatic hypotension, especially at higher doses, may require splitting into 3–4 daily doses
- Check blood pressure before and 45–60 minutes after an individual dose >30 mg until stable
- Patients receiving high doses (up to 120 mg/day if tolerated) may need to be evaluated periodically for effects on the liver

Overdose

- Dizziness, sedation, ataxia, headache, insomnia, restlessness, anxiety, irritability; cardiovascular effects, confusion, respiratory depression, or coma may also occur

Long-Term Use

- Consider periodic evaluation of hepatic function
- MAOIs may lose efficacy long-term

Habit Forming

- Dependence to MAOIs reported but rare

How to Stop

- Generally no need to taper, as the drug wears off slowly over 2–3 weeks

Pharmacokinetics

- Clinical duration of action may be 14–30 days due to irreversible enzyme inhibition



Drug Interactions

- Tramadol (serotonin reuptake inhibitor) may increase the risk of seizures in patients taking an MAOI
- Can cause a fatal “serotonin syndrome” when combined with drugs that block serotonin reuptake, so do not use with a serotonin reuptake inhibitor or for 5 half-lives after stopping the serotonin reuptake inhibitor (see [Table 1](#) after Pearls)
- Hypertensive crisis with headache, intracranial bleeding, and death may result from combining MAOIs with sympathomimetic drugs (e.g., amphetamines, methylphenidate, cocaine, dopamine, epinephrine, norepinephrine)
- Do not combine with another MAOI inhibitor, alcohol, or guanethidine
- Adverse drug reactions can result from combining MAOIs with tricyclic/tetracyclic antidepressants and related compounds, including

carbamazepine, cyclobenzaprine, and mirtazapine, and should be avoided except by experts

- MAOIs in combination with spinal anesthesia may cause combined hypotensive effects
- Combination of MAOIs and CNS depressants may enhance sedation and hypotension

Table 1. Drugs contraindicated due to risk of serotonin syndrome/toxicity

Do Not Use:			
Antidepressants	Drugs of Abuse	Opioids	Other
SSRIs	MDMA (ecstasy)	Meperidine	Non-subcutaneous sumatriptan
SNRIs	Cocaine	Tramadol	Chlorpheniramine
Clomipramine	Methamphetamine	Methadone	Brompheniramine
Imipramine	High-dose or injected amphetamine		Dextromethorphan
St. John's wort			Lumateperone
			Ziprasidone

Other Warnings/Precautions

- Use requires strict adherence to low-tyramine diet (see [Table 2](#) after Pearls)

- Patient and prescriber must be vigilant to potential interactions with any drug, including antihypertensives and over-the-counter cough/cold preparations
- Over-the-counter medications to avoid include cough and cold preparations, including those containing dextromethorphan (serotonin reuptake inhibitor), nasal decongestants (tablets, drops, or spray), hay-fever medications, sinus medications, asthma inhalant medications, anti-appetite medications, weight reducing preparations, “pep” pills (see [Table 3](#) after Pearls)
- Hypoglycemia may occur in diabetic patients receiving insulin or oral antidiabetic agents, although more so with hydrazine MAOIs than with tranylcypromine
- Use cautiously in patients receiving reserpine, anesthetics, disulfiram, metrizamide, anticholinergic agents
- Tranylcypromine is not recommended for use in patients who cannot be monitored closely

Table 2. Dietary guidelines for patients taking MAOIs*

Foods to avoid**	Foods allowed
Dried, aged, smoked, fermented, spoiled, or improperly stored meat, poultry, and fish	Fresh or processed meat, poultry, and fish; properly stored pickled or smoked fish
Broad bean pods	All other vegetables

Foods to avoid**	Foods allowed
Aged cheeses	Processed cheese slices, cottage cheese, ricotta cheese, yogurt, cream cheese
Tap and unpasteurized beer	Canned or bottled beer and alcohol
Marmite	Brewer's and baker's yeast
Banana peel	Bananas, avocados, raspberries
Sauerkraut, kimchee	Peanuts
Soy products/tofu	
Tyramine-containing nutritional supplement	

* Consult up-to-date listing of tyramine content, as most foods now have low tyramine amounts

** Not necessary for 6 mg transdermal or low-dose oral selegiline

Table 3. Drugs that boost norepinephrine: should only be used with caution with MAOIs

Use With Caution:			
Decongestants	Stimulants	Antidepressants with norepinephrine reuptake inhibition	Other

Use With Caution:

Decongestants	Stimulants	Antidepressants with norepinephrine reuptake inhibition	Other
Phenylephrine	Amphetamines	Most tricyclics	Phentermine
Pseudoephedrine	Methylphenidate Cocaine Methamphetamine Modafinil Armodafinil	NRIs NDRI	Local anesthetics containing vasoconstrictors Tapentadol

Do Not Use

- If patient is taking meperidine (pethidine) (serotonin reuptake inhibitor)
- If patient is taking a sympathomimetic agent or taking guanethidine
- If patient is taking another MAOI
- If patient is taking any agent that can inhibit serotonin reuptake (e.g., SSRIs, tramadol, milnacipran, duloxetine, venlafaxine, clomipramine, imipramine, etc.)
- If patient is taking dextromethorphan (serotonin reuptake inhibitor)
- If patient is taking diuretics or other antihypertensives without careful monitoring of blood pressure
- If patient has pheochromocytoma

- If patient has cardiovascular or cerebrovascular disease until cleared by the patient's medical doctor
- If patient is taking a prohibited drug
- If patient is not compliant with a low-tyramine diet
- If there is a proven allergy to tranylcypromine

Special Populations

Renal Impairment

- Use with caution –drug may accumulate in plasma
- May require lower than usual adult dose

Hepatic Impairment

- Tranylcypromine should be used cautiously in patients with history of hepatic impairment or in patients with abnormal liver function tests

Cardiac Impairment

- Contraindicated in patients with cardiac impairment unless cleared by the patient's medical doctor

Elderly

- Initial dose often lower than usual adult dose
- Elderly patients may have greater sensitivity to adverse effects, but many tolerate MAOIs

- Reduction in the risk of suicidality with antidepressants compared to placebo in adults age 65 and older



Children and Adolescents

- Not generally recommended for use in children under age 18
- Carefully weigh the risks and benefits of pharmacological treatment against the risks and benefits of nontreatment with antidepressants and make sure to document this in the patient's chart
- Monitor patients face-to-face regularly, particularly during the first several weeks of treatment
- Use with caution, observing for activation of known or unknown bipolar disorder and/or suicidal ideation, and inform parents or guardians of this risk so they can help observe child or adolescent patients



Pregnancy

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001
- Controlled studies have not been conducted in pregnant women
- Not generally recommended for use during pregnancy, especially during first trimester

- Should evaluate patient for treatment with an antidepressant with a better risk/benefit ratio
- National Pregnancy Registry for Antidepressants: 1-866-961-2388, <https://womensmentalhealth.org/research/pregnancyregistry/antidepressants>

Breast Feeding

- Some drug is found in mother's breast milk
- Effects on infant unknown
- Immediate postpartum period is a high-risk time for depression, especially in women who have had prior depressive episodes, so drug may need to be reinstituted late in the third trimester or shortly after childbirth to prevent a recurrence during the postpartum period
- Should evaluate patient for treatment with an antidepressant with a better risk/benefit ratio

The Art of Psychopharmacology

Potential Advantages

- Melancholia
- Atypical depression
- Severe depression
- Treatment-resistant depression or anxiety disorders

Potential Disadvantages

- Requires strict adherence to dietary restrictions, concomitant drug restrictions
- Patients with cardiac problems or hypertension unless cleared by their medical doctor
- Multiple daily doses unless drug can be converted to once a day

Primary Target Symptoms

- Depressed mood
- Somatic symptoms
- Sleep and eating disturbances
- Psychomotor retardation
- Morbid preoccupation
- Anxiety

Pearls

- Generally less weight gain or sexual dysfunction than with phenelzine
- MAOIs are generally reserved for third- or fourth-line use after SSRIs, SNRIs, and combinations of newer antidepressants have failed
- Patient should be advised not to take any prescription or over-the-counter drugs without consulting their doctor because of possible drug interactions with the MAOI
- Headache is often the first symptom of hypertensive crisis

- The rigid dietary restrictions may reduce compliance (see [Table 2](#) after Pearls)
- Mood disorders can be associated with eating disorders (especially in adolescent females), and tranylcypromine can be used to treat both depression and bulimia
- * Myths about the danger of dietary tyramine can be exaggerated, but prohibitions against concomitant drugs often not followed closely enough
- Orthostatic hypotension, insomnia, and sexual dysfunction are often the most troublesome common side effects
- * MAOIs should be for the expert, especially if combining with agents of potential risk (e.g., stimulants, trazodone, TCAs)
- * MAOIs should not be neglected as therapeutic agents for the treatment-resistant
- For treatment-resistant patients, experts can give a tricyclic/tetracyclic antidepressant other than clomipramine or imipramine simultaneously with an MAOI for patients who fail to respond to numerous other antidepressants
- Use of MAOIs with clomipramine is always prohibited because of the risk of serotonin syndrome and death due to serotonin reuptake inhibition
- Amoxapine may be the preferred tricyclic/tetracyclic antidepressant to combine with an MAOI in heroic cases due to its theoretically protective 5HT_{2A} antagonist properties

- If this option is elected, start the MAOI with the tricyclic/tetracyclic antidepressant simultaneously at low doses after appropriate drug washout, then alternately increase doses of these agents every few days to a week as tolerated
- Although very strict dietary and concomitant drug restrictions must be observed to prevent hypertensive crises and serotonin syndrome, the most common side effects of MAOI and tricyclic/tetracyclic combinations may be weight gain and orthostatic hypotension

Suggested Reading

Gillman K. PsychoTropical Research (PTR). <http://psychotropic.com> .

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Gillman PK . A reassessment of the safety profile of monoamine oxidase inhibitors: elucidating tired old tyramine myths . *J Neural Transm (Vienna)* 2018 ;**125** (11):1707 –17 .

.....

Ricken R , Ulrich S , Schlattmann P , Adli M . Tranylcypromine in mind (part II): review of clinical pharmacology and meta-analysis of controlled studies in depression . *Eur Neuropsychopharmacol* 2017 ;**27** (8):714 –31

.....

Ulrich S , Ricken R , Adli M . Tranylcypromine in mind (part I): review of pharmacology . *Eur Neuropsychopharmacol* 2017 ;**27** (8):697 –13 .

.....

Ulrich S , Ricken R , Buspavanich P , Schlattmann P , Adli M . Efficacy and adverse effects of tranylcypromine and tricyclic antidepressants in the treatment of depression: a systematic review and comprehensive meta-analysis . *J Clin Psychopharmacol* 2020 ;**40** (1):63 –74 .

Trazodone

Therapeutics

Brands

- Desyrel
- Oleptro
- *see index for additional brand names*

Generic?

Yes

Class

- Neuroscience-based Nomenclature: serotonin receptor antagonist (S-MM)
- SARI (serotonin 2 antagonist/reuptake inhibitor); antidepressant; hypnotic

Commonly Prescribed for

(bold for FDA approved)

- **Depression**
- Insomnia (primary and secondary)
- Anxiety

How the Drug Works

- Blocks serotonin 2A receptors potently
- Blocks serotonin reuptake pump (serotonin transporter) less potently

How Long Until It Works

* Onset of therapeutic actions in insomnia are immediate if dosing is correct

- Onset of therapeutic actions in depression usually not immediate, but often delayed 2–4 weeks whether given as an adjunct to another antidepressant or as a monotherapy
- If it is not working within 6–8 weeks for depression, it may require a dosage increase or it may not work at all
- May continue to work for many years to prevent relapse of symptoms in depression and to reduce symptoms of chronic insomnia

If It Works

* For insomnia, use possibly can be indefinite as there is no reliable evidence of tolerance, dependence, or withdrawal, but few long-term studies

- For secondary insomnia, if underlying condition (e.g., depression, anxiety disorder) is in remission, trazodone treatment may be discontinued if insomnia does not reemerge

- The goal of treatment for depression is complete remission of current symptoms of depression as well as prevention of future relapses
- Treatment most often reduces or even eliminates symptoms of depression, but is not a cure since symptoms can recur after medicine stopped
- Continue treatment until all symptoms of depression are gone (remission)
- Once symptoms of depression are gone, continue treating for 1 year for the first episode of depression
- For second and subsequent episodes of depression, treatment may need to be indefinite

If It Doesn't Work

- For insomnia, try escalating doses or switch to another agent
- Many patients have only a partial antidepressant response where some symptoms are improved but others persist (especially insomnia, fatigue, and problems concentrating)
- Other patients may be nonresponders, sometimes called treatment-resistant or treatment-refractory
- Consider increasing dose, switching to another agent, or adding an appropriate augmenting agent for treatment of depression
- Consider psychotherapy

- Consider evaluation for another diagnosis or for a comorbid condition (e.g., medical illness, substance abuse, etc.)
- Some patients may experience apparent lack of consistent efficacy due to activation of latent or underlying bipolar disorder, and require antidepressant discontinuation and a switch to a mood stabilizer

Best Augmenting Combos for Partial Response or Treatment Resistance

- Trazodone is not frequently used as a monotherapy for insomnia, but can be combined with sedative hypnotic benzodiazepines in difficult cases
- Trazodone is most frequently used in depression as an augmenting agent to numerous psychotropic drugs
- Trazodone can not only improve insomnia in depressed patients treated with antidepressants, but can also be an effective booster of antidepressant actions of other antidepressants (use combinations of antidepressants with caution as this may activate bipolar disorder and suicidal ideation)
- Trazodone can also improve insomnia in numerous other psychiatric conditions (e.g., bipolar disorder, schizophrenia, alcohol withdrawal) and be added to numerous other psychotropic drugs (e.g., lithium, mood stabilizers, antipsychotics)

Tests

- None for healthy individuals

Side effects

How Drug Causes Side Effects

- Sedative effects may be due to antihistamine properties
- Blockade of alpha adrenergic 1 receptors may explain dizziness, sedation, and hypotension
- Most side effects are immediate but often go away with time

Notable Side Effects

- Nausea, vomiting, edema, blurred vision, constipation, dry mouth
- Dizziness, sedation, fatigue, headache, incoordination, tremor
- Hypotension, syncope
- Occasional sinus bradycardia (long-term)
- Rare rash



Life-Threatening or Dangerous Side Effects

- Rare priapism
- Rare seizures
- Rare induction of mania
- Rare activation of suicidal ideation and behavior (suicidality) (short-term studies did not show an increase in the risk of suicidality with antidepressants compared to placebo beyond age 24)

Weight Gain



- Reported but not expected

Sedation



- Many experience and/or can be significant in amount

What to Do About Side Effects

- Wait
- Wait
- Wait
- Take larger dose at night to prevent daytime sedation
- Switch to another agent

Best Augmenting Agents for Side Effects

- Most side effects cannot be improved with an augmenting agent
- Activation and agitation may represent the induction of a bipolar state, especially a mixed dysphoric bipolar II condition sometimes associated with suicidal ideation, and require the addition of lithium, a mood stabilizer or an atypical antipsychotic, and/or discontinuation of trazodone

Dosing and use

Usual Dosage Range

- 150–600 mg/day

Dosage Forms

- Tablet 50 mg scored, 100 mg scored, 150 mg, 150 mg with povidone scored, 300 mg with povidone scored

How to Dose

- Depression as a monotherapy: initial 150 mg/day in divided doses; can increase every 3–4 days by 50 mg/day as needed; maximum 400 mg/day (outpatient) or 600 mg/day (inpatient), split into 2 daily doses
- Insomnia: initial 25–50 mg at bedtime; increase as tolerated, usually to 50–100 mg/day, but some patients may require up to full antidepressant dose range
- Augmentation of other antidepressants in the treatment of depression: dose as recommended for insomnia



Dosing Tips

- Start low and go slow
- * Patients can have carryover sedation, ataxia, and intoxicated-like feeling if dosed too aggressively, particularly when initiating dosing

* Do not discontinue trials if ineffective at low doses (<50 mg) as many patients with difficult cases may respond to higher doses (150–300 mg, even up to 600 mg in some cases)

- For relief of daytime anxiety, can give part of the dose in the daytime if not too sedating
- Although use as a monotherapy for depression is usually in divided doses due to its short half-life, use as an adjunct is often effective and best tolerated once daily at bedtime

Overdose

- Rarely lethal; sedation, vomiting, priapism, respiratory arrest, seizure, ECG changes

Long-Term Use

- Safe

Habit Forming

- No

How to Stop

- Taper is prudent to avoid withdrawal effects, but tolerance, dependence, and withdrawal effects have not been reliably demonstrated

Pharmacokinetics

- Metabolized by CYP450 3A4
- Half-life is biphasic; first phase is approximately 3–6 hours; second phase is approximately 5–9 hours



Drug Interactions

- Tramadol increases the risk of seizures in patients taking an antidepressant
- Fluoxetine and other SSRIs may raise trazodone plasma levels
- Trazodone may block the hypotensive effects of some antihypertensive drugs
- Trazodone may increase digoxin or phenytoin concentrations
- Trazodone may interfere with the antihypertensive effects of clonidine
- Generally, do not use with MAOIs, including 14 days after MAOIs are stopped
- Reports of increased and decreased prothrombin time in patients taking warfarin and trazodone



Other Warnings/Precautions

- Possibility of additive effects if trazodone is used with other CNS depressants

- Treatment should be discontinued if prolonged penile erection occurs because of the risk of permanent erectile dysfunction
- Advise patients to seek medical attention immediately if painful erections occur lasting more than 1 hour
- Generally, priapism reverses spontaneously, while penile blood flow and other signs are being monitored, but in urgent cases, local phenylephrine injections or even surgery may be indicated
- Use with caution in patients with history of seizures
- Use with caution in patients with bipolar disorder unless treated with concomitant mood-stabilizing agent
- When treating children, carefully weigh the risks and benefits of pharmacological treatment against the risks and benefits of nontreatment with antidepressants and make sure to document this in the patient's chart
- Distribute the brochures provided by the FDA and the drug companies
- Whenever possible, warn patients and their caregivers about the possibility of activating side effects, and advise them to report such symptoms immediately
- Monitor patients for activation of suicidal ideation, especially children and adolescents

Do Not Use

- If patient is taking an MAOI, but see Pearls